Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

AI platform engineer who **built AgentProbe end-to-end** - the exact system Slack's AI Platform team builds: **AI quality eval frameworks** (5-metric LLM-as-judge engine with CI/CD regression gates), LLM operationalization across multiple providers (Anthropic, OpenAI, Groq, Ollama), and prompt engineering infrastructure. **Claude Code and GitHub Copilot are daily tools** - explicitly in my workflow. **8+ years production Python**; full-stack delivery across 120+ Agile sprints at Entrata (100K+ MAU, 99.5% uptime); TypeScript/JavaScript across 20+ modules. AWS (S3, EC2, RDS, Glue) in production. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

AI Quality Eval Frameworks (LLM-as-Judge) | LLM Operationalization & Inference | Prompt & Context Engineering | CI/CD Pipelines for AI Integration | Python (8+ years production) | RAG Pipelines & Vector Search (ChromaDB) | Multi-Agent Orchestration (LangGraph) | TypeScript / JavaScript / React | FastAPI / REST Platform Layer | AWS (S3, EC2, RDS, Glue) / Docker | Claude Code & GitHub Copilot (daily) | Production at Scale (1M+ records, 100K+ MAU)

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated data with output validation, quality monitoring, and compliance controls.
- Built and delivered **20+ full-stack modules** (Python, TypeScript/JavaScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users** - owned large product areas end-to-end from scoping through production.
- Led Docker-based containerization for **30%+ of microservices**, cutting execution time by **50%**; improved production query performance by **30%** across 200+ SQL queries; maintained **99.5% uptime** for 6+ months.
- Migrated client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents** across 100K+ users; managed EC2, RDS, and Glue.
- Built **5+ internal platform tools** adopted across squads, reducing resolution time by **40%**; drove secure coding standards and peer code review practices across all releases.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and database systems; designed supplementary exercises and drove iterative quality improvement across the course.

PROJECTS

AgentProbe: AI Eval & LLM Operations Platform - Quality Frameworks, CI/CD, Multi-Provider OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD · Claude Code

Built Slack AI Platform's exact system - AI quality eval frameworks, LLM operationalization, and CI/CD pipelines for AI integration - from greenfield to open-source production. **AI quality eval framework**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with per-run regression tracking and **CI/CD gates** - any team can call the eval API to gate deployments, exactly the pattern Slack's AI Platform runs at scale. **LLM operationalization**: multi-provider routing across Anthropic, OpenAI, Groq, Ollama - dynamic selection by task type, latency, and capability at the API layer. **Prompt & context engineering**: context window management, token budget optimization, short vs. long-term memory patterns - the infrastructure layer beneath any AI product feature. **Multi-agent runtime**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) with tool routing and inter-agent state handoffs. **Platform layer**: 15-endpoint FastAPI backend + React observability dashboard. Built with **Claude Code as primary development tool**. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: Domain-Specific NLP Platform - LLMs + FHIR APIs

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

NLP platform routing natural language queries to domain-specific FHIR APIs via LLMs, BERT, and embeddings - HIPAA-compliant RBAC. **~60% retrieval time reduction.** Built for a regulated enterprise domain with strict compliance requirements.
github.com/venkatalakshmiparimala/AI_on_FHIR

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Platform: LangChain, LangGraph, Multi-Agent Orchestration, LLM APIs (Anthropic, OpenAI, Groq, Ollama), AI Quality Eval Frameworks, LLM-as-Judge, CI/CD Eval Gates, RAG Pipelines, ChromaDB, Vector Search, Prompt Engineering, Context Engineering, Token Management

Backend & Infra: FastAPI, Flask, React, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Statistical Modeling

Dev Tools: Claude Code, GitHub Copilot, Cursor, TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum